

09_statistical_analysis

October 11, 2026

COMPSCI 760 Group 9, statistical analysis

Input : merged_3system_with_metrics.csv Output: /intervals_long.csv paired bootstrap CIs on gaps + accent spread /accent_means_ci.csv accent means per system with CIs, ranks, spread /spearman.csv WER vs cascade chrF++ / vs Direct-minus-Cascade gap /tests.csv Kruskal-Wallis, Dunn (Holm), effect sizes /intervals_unk_free.csv gap intervals on the clips where Direct did not output

Usage python statistical_analysis.py # reads merged_3system_with_metrics.csv in the runs folder

Expected columns (fixed): id, sample_id (speaker), primary_accent, wer_norm, chrftp_direct, chrftp_600m, chrftp_33b, direct_has_unk

Conventions * outcome = sentence-level chrF++ (0-100) * gap = first system minus second system (negative => first system scores lower) * default resampling unit = speaker, stratified by accent; clip-level reported alongside * seed 760, 1000 bootstrap iterations, 95% percentile intervals “ “ ”

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[11]: %pip install numpy pandas scip
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Requirement already satisfied: numpy in
/Users/nlh/Downloads/uoa_lectures/CS760/project/cs760-accent-st-robustness-ml-
research/.venv/lib/python3.9/site-packages (2.0.2)
Requirement already satisfied: pandas in
/Users/nlh/Downloads/uoa_lectures/CS760/project/cs760-accent-st-robustness-ml-
research/.venv/lib/python3.9/site-packages (2.3.3)
Requirement already satisfied: scip in
/Users/nlh/Downloads/uoa_lectures/CS760/project/cs760-accent-st-robustness-ml-
research/.venv/lib/python3.9/site-packages (0.1.12)
Requirement already satisfied: tzdata>=2022.7 in
/Users/nlh/Downloads/uoa_lectures/CS760/project/cs760-accent-st-robustness-ml-
research/.venv/lib/python3.9/site-packages (from pandas) (2026.3)
Requirement already satisfied: python-dateutil>=2.8.2 in
/Users/nlh/Downloads/uoa_lectures/CS760/project/cs760-accent-st-robustness-ml-
research/.venv/lib/python3.9/site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
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research/.venv/lib/python3.9/site-packages (from scip) (0.10.9)
Requirement already satisfied: graphviz in
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Requirement already satisfied: dask[distributed]>=2022.01.1 in
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research/.venv/lib/python3.9/site-packages (from scip) (6.0.3)
Requirement already satisfied: Pillow in
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research/.venv/lib/python3.9/site-packages (from scip) (0.60.0)
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Requirement already satisfied: partd>=1.4.0 in
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dask[distributed]>=2022.01.1->scip) (1.4.2)
Requirement already satisfied: packaging>=20.0 in
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Requirement already satisfied: msgpack>=1.0.0 in
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research/.venv/lib/python3.9/site-packages (from importlib-

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metadata>=4.13.0->dask[distributed]>=2022.01.1->scip) (3.23.1)
Requirement already satisfied: MarkupSafe>=2.0 in
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(1.17.0)
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types>=0.3.4->aicsimageio->scip) (25.7)

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 yarl<2.0,>=1.17.0->aiohttp<4->ome-zarr>=0.6.1->aicsimageio->scip) (3.20)
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 research/.venv/lib/python3.9/site-packages (from requests->ome-
 zarr>=0.6.1->aicsimageio->scip) (2026.7.22)

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Requirement already satisfied: aiobotocore~=2.5.0 in
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 research/.venv/lib/python3.9/site-packages (from
 s3fs->fsspec>=2021.09.0->dask[distributed]>=2022.01.1->scip) (2.5.4)

Requirement already satisfied: botocore<1.31.18,>=1.31.17 in
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 (1.31.17)

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Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in
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 >aiobotocore~=2.5.0->s3fs->fsspec>=2021.09.0->dask[distributed]>=2022.01.1-
 >scip) (1.1.0)

Requirement already satisfied: imageio>=2.33 in
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 research/.venv/lib/python3.9/site-packages (from scikit-image->scip) (2.37.2)

Requirement already satisfied: networkx>=2.8 in
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 research/.venv/lib/python3.9/site-packages (from scikit-image->scip) (3.2.1)

Requirement already satisfied: lazy-loader>=0.4 in
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 research/.venv/lib/python3.9/site-packages (from scikit-image->scip) (0.5)

Requirement already satisfied: elementpath<6.0.0,>=5.0.1 in
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 research/.venv/lib/python3.9/site-packages (from xmlschema->aicsimageio->scip)
 (5.0.4)

WARNING: You are using pip version 21.2.4; however, version 26.0.1 is
 available.

You should consider upgrading via the

'/Users/nlh/Downloads/uoa_lectures/CS760/project/cs760-accent-st-robustness-ml-
 research/.venv/bin/python -m pip install --upgrade pip' command.

Note: you may need to restart the kernel to use updated packages.

```
[12]: from __future__ import annotations
```

```

import argparse
import itertools
import sys
import time
from pathlib import Path

import numpy as np
import pandas as pd
from scipy import stats

```

```

[13]: from pathlib import Path

def find_project_root():
    cwd = Path.cwd().resolve()
    for candidate in [cwd, cwd.parent]:
        if (candidate / "data").exists() and (candidate / "runs").exists():
            return candidate
    return cwd

PROJECT_ROOT = find_project_root()

DATA_FILE = PROJECT_ROOT / "runs" / "evaluation_results" /
↳ "merged_3system_with_metrics.csv"

OUTPUT_DIR = (
    PROJECT_ROOT
    / "runs"
    / "statistical_analysis_results"
)

OUTPUT_DIR.mkdir(parents=True, exist_ok=True)

if not DATA_FILE.is_file():
    raise FileNotFoundError(f"Merged CSV not found: {DATA_FILE}")

print("Project root:", PROJECT_ROOT)
print("Input:", DATA_FILE)
print("Output:", OUTPUT_DIR)

```

```

Project root: /Users/nlh/Downloads/uoa_lectures/CS760/project/cs760-accent-st-
robustness-ml-research
Input: /Users/nlh/Downloads/uoa_lectures/CS760/project/cs760-accent-st-
robustness-ml-research/runs/evaluation_results/merged_3system_with_metrics.csv
Output: /Users/nlh/Downloads/uoa_lectures/CS760/project/cs760-accent-st-
robustness-ml-research/runs/statistical_analysis_results

```



```
[14]: SEED, N_BOOT = 760, 1000
SYSTEMS = ["direct", "c600m", "c33b"]
LABEL = {"direct": "Direct", "c600m": "Cascade-600M", "c33b": "Cascade-3.3B"}
GAPS = {"direct_minus_600m": ("direct", "c600m"),
        "direct_minus_33b": ("direct", "c33b"),
        "33b_minus_600m": ("c33b", "c600m")}
GAP_LABEL = {"direct_minus_600m": "Direct minus Cascade-600M",
             "direct_minus_33b": "Direct minus Cascade-3.3B",
             "33b_minus_600m": "Cascade-3.3B minus Cascade-600M"}
PROVIDED_GAP_COLS = {"direct_minus_600m": "gap_direct_600m",
                     "direct_minus_33b": "gap_direct_33b",
                     "33b_minus_600m": "gap_33b_600m"}
OVERALL = "overall"
```

- 1 _____
- 2 1. Loading (columns are fixed to merged_3system_with_metrics.csv)
- 3 _____

```
[15]: COLS = {"id": "id", "speaker": "sample_id", "accent": "primary_accent", "wer": "wer_norm",
             "direct": "chrhpp_direct", "c600m": "chrhpp_600m", "c33b": "chrhpp_33b",
             "unk": "direct_has_unk"}

def read_csv(path):
    if not Path(path).exists():
        sys.exit(f"File not found: {path}\nPut the CSV in the folder you run the command from, or pass --data <path>.")
    return pd.read_csv(path, encoding="utf-8-sig") # utf-8-sig also copes with a BOM

def prepare(raw):
    missing = [c for c in COLS.values() if c not in raw.columns]
    if missing:
        sys.exit(f"Missing expected column(s): {missing}\nColumns in the file: {list(raw.columns)}")
    if raw[COLS["speaker"]].isna().any():
        sys.exit(f"{int(raw[COLS['speaker']].isna().sum())} rows have an empty {COLS['speaker']}; fix the file first.")
    d = pd.DataFrame({
        "id": raw[COLS["id"]].astype(str),
        "accent": raw[COLS["accent"]].astype(str).str.strip(),
        "speaker": raw[COLS["speaker"]].astype(str),
```

```

        "wer": pd.to_numeric(raw[COLS["wer"]], errors="coerce"),
        "unk": raw[COLS["unk"]].astype(str).str.strip().str.lower().
↪isin(["true", "1", "1.0", "yes"]),
    })
    for s in SYSTEMS:
        d[s] = pd.to_numeric(raw[COLS[s]], errors="coerce")
    for g, (a, b) in GAPS.items():
        d[g] = d[a] - d[b]
        pc = PROVIDED_GAP_COLS[g]
        if pc in raw.columns:
            diff = (pd.to_numeric(raw[pc], errors="coerce") - d[g]).abs().max()
            if diff > 1e-6:
                print(f" WARNING: provided column {pc} differs from {a}-{b} "
↪f"(max abs diff {diff:.4g}). Using recomputed values_
↪(first minus second).")
        n0 = len(d)
        d = d.dropna(subset=SYSTEMS).reset_index(drop=True)
        if len(d) < n0:
            print(f" WARNING: dropped {n0 - len(d)} rows with missing chrF++.")
        if d["id"].duplicated().any():
            sys.exit("Duplicate ids in the merged file - fix the merge before_
↪analysing.")
        multi = d.groupby("speaker")["accent"].nunique()
        if (multi > 1).any():
            print(f" WARNING: {(multi > 1).sum()} speaker ids appear under more_
↪than one accent; "
                "they are treated as separate clusters within each accent.")
    return d

```

4

5 2. Cluster bootstrap (stratified by accent, one shared set of resamples)

6

```

[16]: class StratifiedBoot:
        """Resample clusters (speakers, or single clips) with replacement WITHIN_
↪each accent.
        The same resampled clusters are used for every value column, so every_
↪statistic
        (system means, paired gaps, spreads) is paired across systems."""
        def __init__(self, df, value_cols, unit, B, seed):

```

```

        self.cols, self.accents = list(value_cols), sorted(df["accent"].
↪unique())
        rng = np.random.default_rng(seed)
        self.num, self.den = {}, {}
        for a in self.accents:
            d = df[df["accent"] == a]
            key = d["speaker"].to_numpy() if unit == "speaker" else np.
↪arange(len(d))
            codes, _ = pd.factorize(key)
            G = codes.max() + 1
            S = np.zeros((G, len(self.cols)))
            np.add.at(S, codes, d[self.cols].to_numpy())
            n = np.bincount(codes, minlength=G).astype(float)
            idx = rng.integers(0, G, size=(B, G))
            self.num[a] = S[idx].sum(axis=1)                # (B, K) resampled sums
            self.den[a] = n[idx].sum(axis=1)                # (B,)  resampled clip
↪counts

        def means(self, accent=None):                      # -> (B, K)
            if accent is None:
                return sum(self.num.values()) / sum(self.den.values())[:, None]
            return self.num[accent] / self.den[accent][:, None]

        def spread(self):                                  # max - min of accent
↪means, (B, K)
            M = np.stack([self.means(a) for a in self.accents])
            return M.max(axis=0) - M.min(axis=0)

def _ci(x):
    lo, hi = np.nanpercentile(x, [2.5, 97.5])
    return lo, hi, np.nanstd(x, ddof=1)

def _row(statistic, name, stratum, unit, d, est, draws, **extra):
    lo, hi, se = _ci(draws)
    return dict(statistic=statistic, comparison=name, stratum=stratum,
↪unit=unit,
                n_clips=len(d), n_speakers=d["speaker"].nunique(), estimate=est,
                ci_low=lo, ci_high=hi, boot_se=se,
                excludes_zero=(bool(lo > 0 or hi < 0)
↪if statistic in ("mean_gap",
↪"spread_difference") else np.nan), **extra)

def bootstrap_tables(df, unit, B, seed):

```

```

cols = SYSTEMS + list(GAPS)
bt = StratifiedBoot(df, cols, unit, B, seed)
k = {c: i for i, c in enumerate(cols)}
iv, am = [], []
for stratum in [OVERALL] + bt.accents:
    d = df if stratum == OVERALL else df[df["accent"] == stratum]
    M = bt.means(None if stratum == OVERALL else stratum)
    obs = d[cols].mean()
    for g in GAPS:
        iv.append(_row("mean_gap", GAP_LABEL[g], stratum, unit, d, obs[g],
↪M[:, k[g]]))
    if stratum != OVERALL:
        for s in SYSTEMS:
            r = _row("accent_mean", LABEL[s], stratum, unit, d, obs[s], M[
↪, k[s]])
            r["system"] = r.pop("comparison")
            am.append(r)
    # between-accent spread (max - min of the 7 accent means)
    am_df = pd.DataFrame(am)
    SP = bt.spread()
    obs_means = df.groupby("accent")[cols].mean()
    spread_obs = obs_means.max() - obs_means.min()
    for s in SYSTEMS:
        hi_a, lo_a = obs_means[s].idxmax(), obs_means[s].idxmin()
        r = _row("spread_max_minus_min", f"spread_{LABEL[s]}", f"{len(bt.
↪accents)} accents",
            unit, df, spread_obs[s], SP[:, k[s]])
        iv.append(r)
        r2 = dict(r, system=LABEL[s], stratum=f"max: {hi_a} / min: {lo_a}")
        r2.pop("comparison")
        am_df = pd.concat([am_df, pd.DataFrame([r2])], ignore_index=True)
    for s1, s2 in [("direct", "c600m"), ("direct", "c33b"), ("c33b", "c600m")]:
        r = _row("spread_difference", f"spread {LABEL[s1]} minus {LABEL[s2]}",
            f"{len(bt.accents)} accents", unit, df, spread_obs[s1] -
↪spread_obs[s2],
            SP[:, k[s1]] - SP[:, k[s2]])
        am_df = pd.concat([am_df, pd.DataFrame([dict(r, system=r.
↪pop("comparison"))])],
            ignore_index=True)
    # rank of each accent within each system (1 = highest mean)
    m = am_df["statistic"] == "accent_mean"
    am_df["rank_within_system"] = np.nan
    am_df.loc[m, "rank_within_system"] = (am_df[m].
↪groupby(["system"])["estimate"]
            .rank(ascending=False, method="min"))
    am_df = am_df.rename(columns={"stratum": "accent"})

```

```

am_df = am_df[["statistic", "system", "accent", "unit"] +
               [c for c in am_df.columns if c not in ("statistic", "system", "u
↪"accent", "unit")]]
return pd.DataFrame(iv), am_df

```

7

8 3. Spearman with speaker-cluster (or clip) bootstrap CI

9

```

[17]: class Clusters:
        """Fast row-index resampler: speakers (or clips) with replacement within
        ↪each accent."""

        def __init__(self, d, unit):
            self.parts = []
            for _, da in d.groupby("accent", sort=True):
                pos = da.index.to_numpy()
                key = da["speaker"].to_numpy() if unit == "speaker" else pos
                codes, _ = pd.factorize(key)
                order = np.argsort(codes, kind="stable")
                lens = np.bincount(codes)
                self.parts.append((pos[order], np.cumsum(lens) - lens, lens))

        def draw(self, rng):
            out = []
            for ordpos, starts, lens in self.parts:
                G = len(lens)
                sp = rng.integers(0, G, G)
                l, s = lens[sp], starts[sp]
                offs = np.repeat(s - (np.cumsum(l) - 1), 1) + np.arange(l.sum())
                out.append(ordpos[offs])
            return np.concatenate(out)

        def _spearman_ranked(rx, y):
            ry = stats.rankdata(y)
            rx_c, ry_c = rx - rx.mean(), ry - ry.mean()
            den = np.sqrt((rx_c ** 2).sum() * (ry_c ** 2).sum())
            return (rx_c * ry_c).sum() / den if den > 0 else np.nan

        def spearman_table(df, unit, B, seed):
            df = df.dropna(subset=["wer"]).reset_index(drop=True)

```

```

ys = [("c600m", "Cascade-600M sentence chrF++"), ("c33b", "Cascade-3.3B_
↪sentence chrF++"),
      ("direct_minus_600m", "gap Direct minus Cascade-600M"),
      ("direct_minus_33b", "gap Direct minus Cascade-3.3B")]
wer = df["wer"].to_numpy()
rows = []
for stratum in [OVERALL] + sorted(df["accent"].unique()):
    d = df if stratum == OVERALL else df[df["accent"] == stratum]
    cl, rng = Clusters(d, unit), np.random.default_rng(seed)
    boots = np.empty((B, len(ys)))
    for b in range(B):
        rows_b = cl.draw(rng)
        rx = stats.rankdata(wer[rows_b])
        for j, (y, _) in enumerate(ys):
            boots[b, j] = _spearman_ranked(rx, df[y].to_numpy()[rows_b])
    for j, (y, lab) in enumerate(ys):
        rho = stats.spearmanr(d["wer"], d[y])[0]
        lo, hi = np.nanpercentile(boots[:, j], [2.5, 97.5])
        rows.append(dict(x="normalised WER", y=lab, stratum=stratum,
↪unit=unit,
                        n_clips=len(d), n_speakers=d["speaker"].nunique(),
                        rho=rho, ci_low=lo, ci_high=hi,
                        excludes_zero=bool(lo > 0 or hi < 0)))
return pd.DataFrame(rows)

```

10

11 4. Kruskal-Wallis, Dunn (Holm), effect sizes

12

```

[18]: def holm(p):
    p = np.asarray(p, float)
    m, order, adj, run = len(p), np.argsort(p), np.empty(len(p)), 0.0
    for rank, i in enumerate(order):
        run = max(run, (m - rank) * p[i])
        adj[i] = min(1.0, run)
    return adj

def cliffs_delta(a, b):
    """P(a>b) - P(a<b). Identical to the Mann-Whitney rank-biserial correlation.
    ↪"""
    r = stats.rankdata(np.concatenate([a, b]))
    u1 = r[:len(a)].sum() - len(a) * (len(a) + 1) / 2
    return 2 * u1 / (len(a) * len(b)) - 1

```

```

def paired_rank_biserial(x):
    x = np.asarray(x, float)
    x = x[x != 0]
    if len(x) == 0:
        return np.nan
    r = stats.rankdata(np.abs(x))
    wp, wm = r[x > 0].sum(), r[x < 0].sum()
    return (wp - wm) / (wp + wm)

def dunn(groups):
    names = list(groups)
    allv = np.concatenate([groups[n] for n in names])
    N = len(allv)
    ranks = stats.rankdata(allv)
    _, t = np.unique(allv, return_counts=True)
    sigma2 = N * (N + 1) / 12 - (t ** 3 - t).sum() / (12 * (N - 1))
    mr, n_i, start = {}, {}, 0
    for n in names:
        L = len(groups[n])
        mr[n], n_i[n] = ranks[start:start + L].mean(), L
        start += L
    out = []
    for a, b in itertools.combinations(names, 2):
        z = (mr[a] - mr[b]) / np.sqrt(sigma2 * (1 / n_i[a] + 1 / n_i[b]))
        out.append(dict(a=a, b=b, z=z, p=2 * stats.norm.sf(abs(z)),
                        delta=cliffs_delta(groups[a], groups[b]), n_a=n_i[a],
↪n_b=n_i[b]))
    for r, ph in zip(out, holm([r["p"] for r in out])):
        r["p_holm"] = ph
    return out

def tests_table(df, accent_means):
    spk = df.groupby(["accent", "speaker"], as_index=False)[SYSTEMS +
↪list(GAPS)].mean()
    rows = []
    targets = [(g, GAP_LABEL[g]) for g in GAPS] + [(s, f"{LABEL[s]} sentence_
↪chrF++") for s in SYSTEMS]
    for level, data in [("clip", df), ("speaker", spk)]:
        for col, lab in targets:
            groups = {a: x[col].to_numpy() for a, x in data.groupby("accent")}
            H, p = stats.kruskal(*groups.values())
            N = sum(len(v) for v in groups.values())

```

```

        rows.append(dict(test="kruskal_wallis", target=lab, level=level,
↪stratum_a="all accents",
                                stratum_b="", n_a=N, n_b=np.nan, statistic=H,
↪df=len(groups) - 1, p=p,
                                p_holm=np.nan, effect_name="epsilon_squared",
                                effect_size=H * (N + 1) / (N ** 2 - 1)))
    if col in GAPS:
        for r in dunn(groups):
            rows.append(dict(test="dunn_holm", target=lab, level=level,
↪stratum_a=r["a"],
                                stratum_b=r["b"], n_a=r["n_a"],
↪n_b=r["n_b"], statistic=r["z"],
                                df=np.nan, p=r["p"], p_holm=r["p_holm"],
                                effect_name="cliffs_delta(a vs b)",
↪effect_size=r["delta"]))
        for stratum, x in [(OVERALL, data)] + list(data.
↪groupby("accent")):
            v = x[col].to_numpy()
            try:
                W, pw = stats.wilcoxon(v[v != 0])
            except ValueError:
                W, pw = np.nan, np.nan
            rows.append(dict(test="wilcoxon_signed_rank", target=lab,
↪level=level,
                                stratum_a=stratum, stratum_b="",
↪n_a=len(v), n_b=np.nan,
                                statistic=W, df=np.nan, p=pw, p_holm=np.
↪nan,
                                effect_name="rank_biserial_paired",
↪effect_size=paired_rank_biserial(v))
            # does the accent ordering agree across systems? (7 accent means per
↪system)
            m = accent_means[(accent_means["statistic"] == "accent_mean") &
↪(accent_means["unit"] == "speaker")]
            piv = m.pivot(index="accent", columns="system", values="estimate")
            for s1, s2 in itertools.combinations(SYSTEMS, 2):
                rho = stats.spearmanr(piv[LABEL[s1]], piv[LABEL[s2]])[0]
                same = bool((piv[LABEL[s1]].rank() == piv[LABEL[s2]].rank()).all())
                rows.append(dict(test="accent_order_spearman", target=f"{LABEL[s1]} vs
↪{LABEL[s2]}", level="accent means",
                                stratum_a="7 accents", stratum_b="identical order" if
↪same else "order differs",
                                n_a=len(piv), n_b=np.nan, statistic=rho, df=np.nan,
↪p=np.nan, p_holm=np.nan,
                                effect_name="spearman_rho", effect_size=rho))
    return pd.DataFrame(rows)

```


13

14 5. main

15

```
[19]: def main():
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.
    ↪RawDescriptionHelpFormatter)
    ap.add_argument("--data", default=DATA_FILE, help=f"path to the merged CSV_
    ↪(default merged_3system_with_metrics.csv)")
    ap.add_argument("--out", default=OUTPUT_DIR, help="output folder (default_
    ↪statistical_analysis_results)")
    ap.add_argument("--B", type=int, default=N_BOOT, help="bootstrap iterations_
    ↪(default 1000)")
    ap.add_argument("--seed", type=int, default=SEED, help="random seed_
    ↪(default 760)")
    args = ap.parse_args()

    raw = read_csv(args.data)
    df = prepare(raw)

    out = Path(args.out)
    out.mkdir(parents=True, exist_ok=True)
    t0 = time.time()
    iv_s, am_s = bootstrap_tables(df, "speaker", args.B, args.seed)
    iv_c, am_c = bootstrap_tables(df, "clip", args.B, args.seed)
    iv, am = pd.concat([iv_s, iv_c], ignore_index=True), pd.concat([am_s,
    ↪am_c], ignore_index=True)
    print(f"\nbootstrap CIs done ({time.time() - t0:.0f}s)")
    sp = pd.concat([spearman_table(df, u, args.B, args.seed) for u in
    ↪("speaker", "clip")], ignore_index=True)
    print(f"spearman done ({time.time() - t0:.0f}s)")
    tests = tests_table(df, am)
    print(f"tests done ({time.time() - t0:.0f}s)")

    iv.to_csv(out / "intervals_long.csv", index=False)
    am.to_csv(out / "accent_means_ci.csv", index=False)
    sp.to_csv(out / "spearman.csv", index=False)
    tests.to_csv(out / "tests.csv", index=False)
    free = df[~df["unk"]].reset_index(drop=True)
    ivf, _ = bootstrap_tables(free, "speaker", args.B, args.seed)
    ivf = ivf[ivf["comparison"].str.contains("Direct")].assign(subset="unk-free_
    ↪clips", n_unk_dropped=int(df["unk"].sum()))
    ivf.to_csv(out / "intervals_unk_free.csv", index=False)
```

```

print(f"unk sensitivity: {int(df['unk'].sum())} Direct <unk> clips excluded_
↳from all systems")

print("\nOverall gaps (speaker-clustered 95% CI):")
show = iv[(iv["unit"] == "speaker") & (iv["stratum"] == OVERALL)]
for _, r in show.iterrows():
    print(f"  {r['comparison']:34s} {r['estimate']:7.2f}  [{r['ci_low']:.
↳2f}, {r['ci_high']:.2f}]")
print(f"\nWrote files to {out.resolve()}")

```

```

[20]: sys.argv = ["statistical_analysis.py"]
      main()

```

```

bootstrap CIs done (0s)
spearman done (7s)
tests done (7s)
unk sensitivity: 508 Direct <unk> clips excluded from all systems

```

```

Overall gaps (speaker-clustered 95% CI):
Direct minus Cascade-600M          7.00  [6.45, 7.60]
Direct minus Cascade-3.3B          4.99  [4.39, 5.70]
Cascade-3.3B minus Cascade-600M    2.01  [1.61, 2.42]

```

```

Wrote files to /Users/nlh/Downloads/uoal_lectures/CS760/project/cs760-accent-st-
robustness-ml-research/runs/statistical_analysis_results

```